

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (EC) Examination

May 2014

EC207 Control Systems

Date: 05.05.2014, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. Use Graph Papers.

SECTION - I

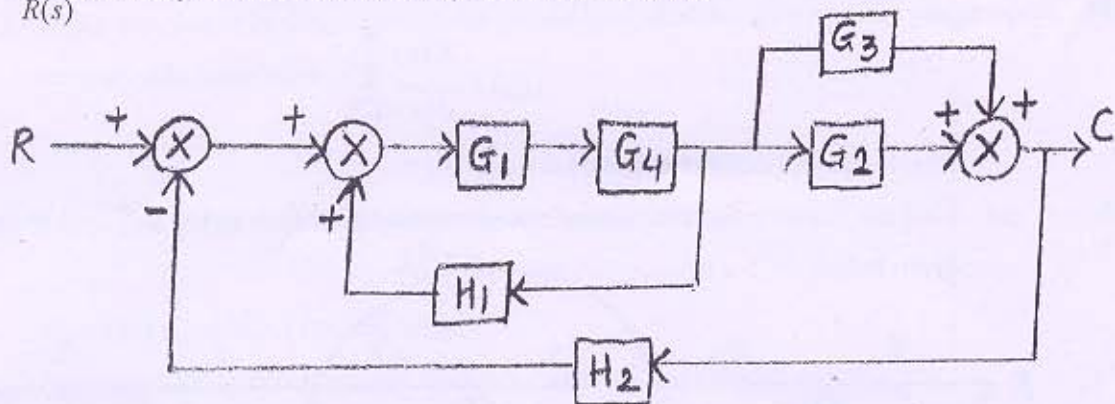
Q - 1 (a) With the help of proper example discuss the difference between open loop and closed loop control systems. [04]

(b) Define following terms. [03]

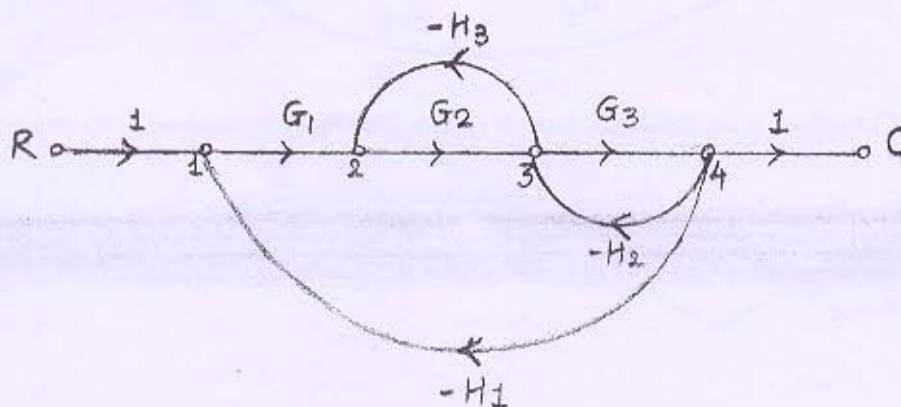
- 1) Maximum Peak overshoot
- 2) Rise Time
- 3) Settling Time

Q - 2 (a) Using block diagram reduction techniques determine closed loop transfer function [07]

$\frac{C(s)}{R(s)}$ of the system whose block diagram is shown below.

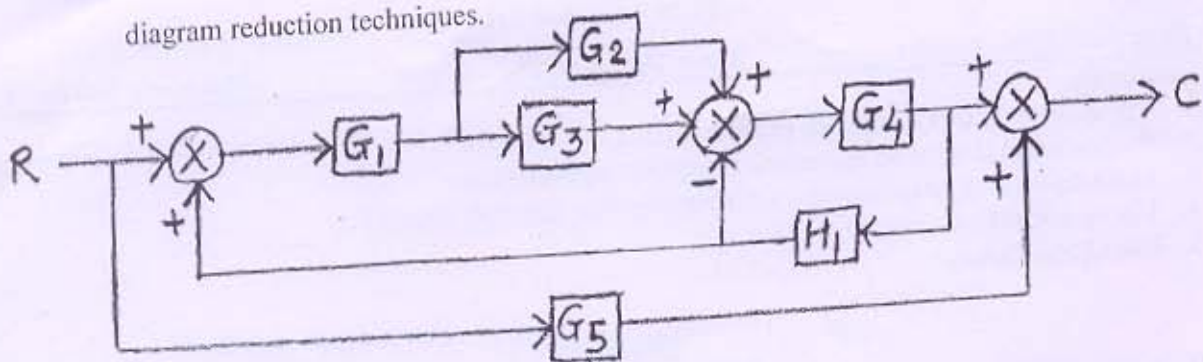


(b) Find the Transfer function for the system whose signal flow graph is shown in figure given below. [07]



OR

- Q - 2 (a) Evaluate the transfer function $\frac{C(s)}{R(s)}$ for the system shown in figure below, using block diagram reduction techniques. [08]



[06]

- (b) A system is described by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -12 & -7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$y = \begin{bmatrix} -10 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + u(t)$$

Test its state controllability and observability.

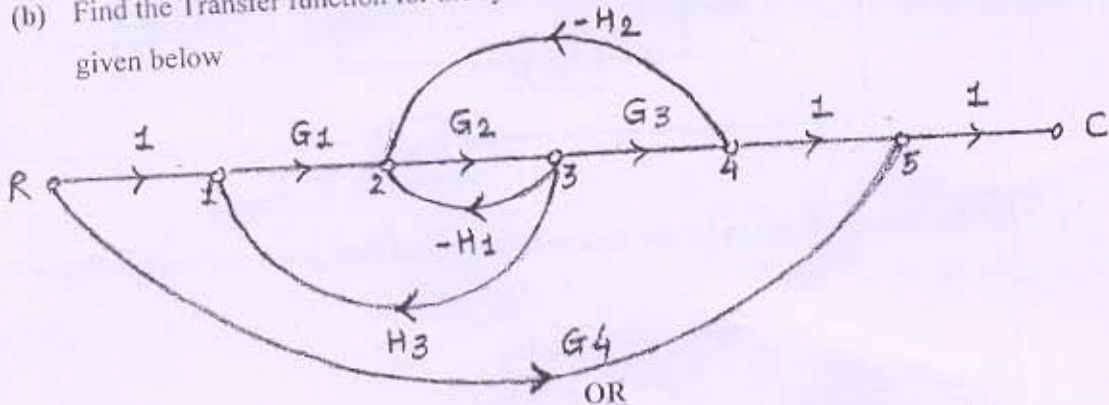
[07]

- Q - 3 (a) A unity feedback control system has an open loop transfer function of

$$G(s) = \frac{K(s + \frac{4}{3})}{s^2(s + 12)}$$

Find the Root Locus as K is varied from 0 to ∞ .

- (b) Find the Transfer function for the system whose signal flow graph is shown in figure [07]



OR

- Q - 3 (a) Consider a unity feedback control system with an open-loop transfer function

$$G(s) = \frac{K}{s(s+3)(s^2+2s+2)}$$

Draw the Root Locus of the system with the gain K as a variable.

[07]

- (b) Prove the following torque equations for the shafts of Gear Trains.

[07]

$$J_{\text{leg}} \ddot{\theta}_1 + f_{\text{leg}} \dot{\theta}_1 + \left(\frac{N_1}{N_2} \right) T_L = T_M$$

SECTION - II

- Q - 4 (a) Sketch the asymptotic bode plot for the transfer function given below. Determine gain cross over frequency, phase cross over frequency, Gain Margin, Phase Margin and stability of the system. [07]

$$G(s)H(s) = \frac{1}{s(s+1)(0.1s+1)}$$

- Q - 5 (a) A unity feedback control system is characterized by the open loop transfer function [07]

$$G(s) = \frac{K(s+13)}{s(s+3)(s+7)}$$

Using Routh Criterion, Calculate the range of values of K for the system to be stable.

- (b) A unity feedback control system has [07]

$$G(s)H(s) = \frac{100e^{-0.1s}}{(s+10)^2}$$

Draw the bode plot.

OR

- Q - 5 (a) Using Nyquist criterion, examine the closed-loop stability of a system whose open-loop transfer function is [08]

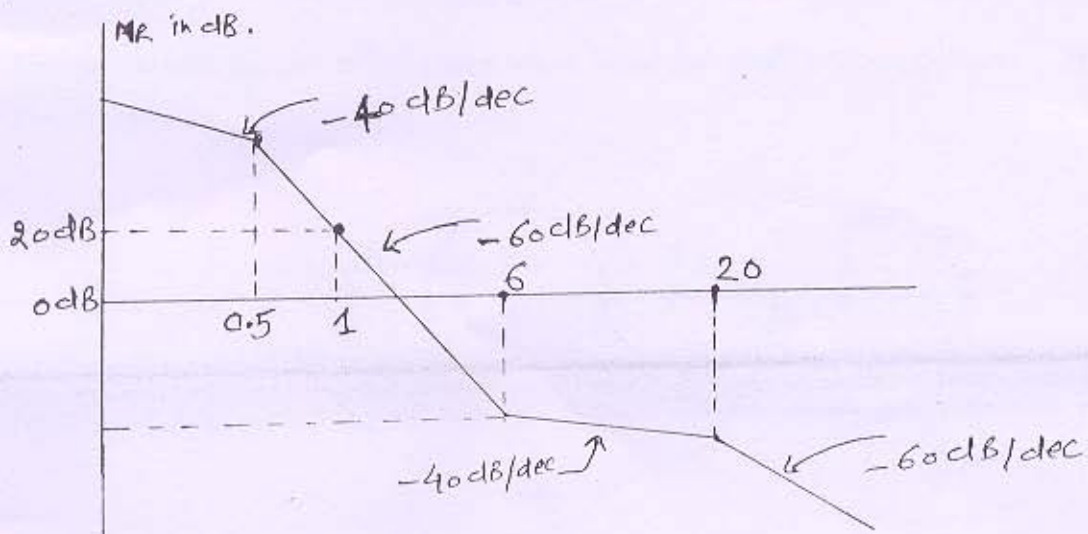
$$G(s)H(s) = \frac{2500}{(s+300)}$$

- (b) Determine the stability of a system whose characteristics equation is [06]

$$s^5 + 2s^4 + 24s^3 + 48s^2 - 25s - 50 = 0$$

Also find the roots of this equation.

- Q - 6 (a) Determine the transfer function from the magnitude plot shown below. [06]



(b) Sketch the Nyquist plot for a system with

[08]

$$G(s)H(s) = \frac{1}{(s+3)(s+2)}$$

Comment on stability.

OR

Q - 6 (a) A second order system has unity feedback and open loop transfer function

[06]

$$G(s) = \frac{500}{s(s+15)}$$

- 1) Obtain characteristic equation.
 - 2) What is damping ratio and natural frequency?
 - 3) Calculate peak time and settling time.
 - 4) Sketch the transient response for unit step input.
- (b) Draw the mechanical network for the mechanical system shown in Figure, Obtain an [08]
analogous electrical circuit based on force- current analogy.

